

AnimatedSwitcher Documentation (Flutter)

Overview

`AnimatedSwitcher` is a Flutter widget that automatically animates between two widgets when its `child` changes. It is commonly used for onboarding screens, step-based UI, and smooth content transitions.

Example Use Case

In this example, `AnimatedSwitcher` is used to animate onboarding images when `currentSteps` changes.

Animation effects used: - Fade animation - Slide animation (right → left)

Source Code Example

```
AnimatedSwitcher(  
  duration: const Duration(milliseconds: 500),  
  transitionBuilder: (child, animation) {  
    return FadeTransition(  
      opacity: animation,  
      child: SlideTransition(  
        position: Tween<Offset>(  
          begin: const Offset(1, 0),  
          end: Offset.zero,  
        ).animate(animation),  
        child: child,  
      ),  
    ),  
  );  
},  
child: Container(  
  key: ValueKey(currentSteps),  
  decoration: BoxDecoration(  
    borderRadius: BorderRadius.circular(18.r),  
  ),  
  child: Image.asset(  
    onboardingItems[currentSteps].image,  
    fit: BoxFit.contain,  
    height: 325.h,  
    width: 345.w,
```

```
    ),  
    ),  
)
```

Property Breakdown

AnimatedSwitcher

Automatically animates when its child widget changes.

duration

Controls how long the animation lasts.

```
Duration(milliseconds: 500)
```

✓ Smooth and fast transition

transitionBuilder

Defines how the old widget exits and the new widget enters.

In this example: - `FadeTransition` controls opacity - `SlideTransition` controls horizontal movement

FadeTransition

Gradually changes opacity from 0 → 1 (enter) and 1 → 0 (exit).

SlideTransition

Moves widget from right to its normal position.

```
begin: Offset(1, 0)  
end: Offset.zero
```

child

The widget being animated. Here it is a `Container` holding an image.

ValueKey(currentSteps)

Keys are **required** for `AnimatedSwitcher`.

✓ Helps Flutter detect widget changes ✓ Triggers animation when step changes

Image.asset

Loads onboarding images dynamically based on `currentSteps`.

Best Practices

- Always use a `Key`
 - Keep animations between 300–600ms
 - Avoid heavy widgets inside transitions
 - Ideal for onboarding & step-based UI
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Summary

This `AnimatedSwitcher` implementation provides a clean, smooth, and professional animation experience. It enhances user engagement and improves UI quality significantly.